

Assessment 03 – Sequence Analysis & GC/AT Content

Course Module: Lecture 05 & Lecture 06 | Bioinformatics & Computational Biology

Subject: Sequence Analysis

Target Gene: TP53 & HBB Transcripts

Submission Status: Complete Solutions

Q1. Sequence Analysis

What is sequence analysis? Explain its importance in bioinformatics and discuss any four of its major applications.

Definition: Sequence analysis is the computational process of subjecting biological sequences (DNA, RNA, or protein/amino acid sequences) to characterization, alignment, pattern recognition, and structural/functional annotation. It aims to determine the biological features, function, structure, and evolutionary origin of a macromolecule.

Importance in Bioinformatics: Sequence analysis forms the cornerstone of computational biology. Because biological sequence data is generated at an exponential rate due to Next-Generation Sequencing (NGS) technologies, sequence analysis allows researchers to decode genomic language, predict gene function without expensive wet-lab experimentation, identify disease mutations, and trace species evolutionary lineages.

Four Major Applications:

1. **Sequence Alignment & Homology Searching (e.g., BLAST):** Compares unknown sequences against database repositories to identify evolutionary conservation, paralogs, and orthologs across organisms.
2. **Gene Prediction & Genome Annotation:** Identifies structural genetic components such as open reading frames (ORFs), exons, introns, transcription factor binding sites, and promoters in raw genomic assemblies.
3. **Phylogenetic & Evolutionary Analysis:** Constructs phylogenetic trees based on sequence divergence, enabling researchers to infer evolutionary pathways, speciation events, and ancestral relationships.
4. **Variant Calling & Disease Diagnostics:** Identifies single nucleotide polymorphisms (SNPs), insertion/deletions (indels), and pathogenic somatic/germline mutations relevant to oncology and genetic disorders.

Q2. GC and AT Content

Define GC content and AT content. Why are these parameters important in genome analysis and molecular biology?

GC Content: The percentage of nitrogenous bases in a nucleic acid molecule that are either Guanine (**G**) or Cytosine (**C**).

$$GC\ Content\ (\%) = [(G + C) / (A + T + G + C)] \times 100\%$$

AT Content: The percentage of nitrogenous bases in a nucleic acid molecule that are either Adenine (**A**) or Thymine (**T**) / Uracil (**U**).

$$AT\ Content\ (\%) = [(A + T) / (A + T + G + C)] \times 100\% = 100\% - GC\ Content\ (\%)$$

Biological & Genomic Importance:

- **Thermal Stability & DNA Melting (T_m):** G-C base pairs form 3 hydrogen bonds and exhibit stronger base-stacking interactions compared to A-T base pairs (2 hydrogen bonds). Higher GC content increases denaturation temperature (T_m).
- **PCR & Primer Design:** Primers with balanced GC content (40%–60%) ensure optimal hybridization temperatures and prevent non-specific binding or hairpin loop secondary structures.
- **Identification of Genomic Islands & HGT:** Horizontal Gene Transfer (HGT) from foreign bacterial or viral DNA often results in genomic regions with distinct GC content compared to the host genome background.
- **Taxonomic Classification:** Bacterial species exhibit distinct GC distributions (ranging from 20% to >70%), serving as a taxonomic and evolutionary signature.

Q3. Scientific Problem — Homo sapiens TP53 mRNA (NM_000546.6)

The DNA sequence was obtained from Homo sapiens TP53 mRNA (RefSeq: NM_000546.6). Show all calculations clearly.

Parameter / Base	Count / Formula	Value
Adenine (A)	N_A	661 bp
Thymine (T)	N_T	532 bp
Guanine (G)	N_G	705 bp
Cytosine (C)	N_C	688 bp
Total Sequence Length	$L = A + T + G + C$	2,586 bp
Total GC Base Count	$G + C = 705 + 688$	1,393 bp
Total AT Base Count	$A + T = 661 + 532$	1,193 bp
GC Content (%)	$(1393 / 2586) \times 100\%$	53.87%
AT Content (%)	$(1193 / 2586) \times 100\%$	46.13%

Calculation Verification:
 $GC\% + AT\% = 53.87\% + 46.13\% = 100.00\%$
 $Total\ Bases = 661 + 532 + 705 + 688 = 2,586\ bp$

Q4. Data Interpretation

Two DNA sequences have the following GC contents:

- **Sequence A:** 35% GC (65% AT)
- **Sequence B:** 68% GC (32% AT)

1. Which sequence is expected to be more thermally stable?

Sequence B (68% GC) is significantly more thermally stable than Sequence A (35% GC).

2. Explain your answer based on the chemical properties of DNA:

Guanine (**G**) and Cytosine (**C**) form **three hydrogen bonds** ($\text{G} \equiv \text{C}$), whereas Adenine (**A**) and Thymine (**T**) form only **two hydrogen bonds** ($\text{A} = \text{T}$). In addition to the extra hydrogen bond, G-C base pairs participate in significantly stronger **π - π base-stacking interactions** along the DNA double helix. Consequently, Sequence B requires higher thermal kinetic energy to disrupt these bonds and melt the double helix into single strands.

3. Which sequence is more likely to belong to a thermophilic organism? Justify your answer:

Sequence B (68% GC) is far more likely to belong to a thermophilic organism. Thermophiles live in high-temperature environments (e.g., thermal vents, hot springs above 60°C–80°C). High genomic GC content provides higher thermal stability, protecting genomic DNA against heat-induced denaturation and maintaining structural integrity under hyperthermic physiological conditions.

Q5. Practical Activity — NCBI Nucleotide Sequence Retrieval

Target Gene: *Homo sapiens* Hemoglobin Subunit Beta (*HBB*), mRNA

NCBI Accession ID: NM_000518.5

FASTA Sequence:

>NM_000518.5 Homo sapiens hemoglobin subunit beta (HBB), mRNA
ACATTTGCTTCTGACACAACCTGTGTTCACTAGCAACCTCAAACAGACACCATGGTGCATCTGACTCCTGA
GGAGAAGTCTGCCGTTACTGCCCTGTGGGGCAAGGTGAACGTGGATGAAGTTGGTGGTGAAGCCCTGGGC
AGGCTGCTGGTGGTCTACCCCTGGACCCAGAGGTTCTTTGAGTCCTTTGGGGATCTGTCCACTCCTGATG
CTGTTATGGGCAACCTAAGGTGAAGGCTCATGGCAAGAAAGTGCTCGGTGCCTTTAGTGATGGCCTGGC
TCACCTGGACAACCTCAAGGGCACCTTTGCCACACTGAGTGAGCTGCACTGTGACAAGCTGCACGTGGAT
CCTGAGAACTTCAGGCTCCTGGGCAACGTGCTGGTCTGTGTGCTGCCCCATCACTTTGGCAAAGAATTCA
CCCCACCAAGTGCAGGCTGCCTATCAGAAAGTGGTGGCTGGTGTGCTAATGCCCTGGCCCCACAAGTATCA
CTAAGCTCGCTTTCTTGCTGTCCAATTTCTATTAAAGGTTCTTTGTTCCCTAAGTCCAACACTACTAAACT
GGGGGATATTATGAAGGGCCTTGAGCATCTGGATTCTGCCTAATAAAAAACATTTATTTTCATTGCAA

Parameter	Calculation Details	Result
Accession ID	NCBI RefSeq Database	NM_000518.5
Sequence Length	Total count of nucleotides	628 bp
Nucleotide Breakdown	A = 142, T = 128, G = 181, C = 177	Total = 628 bp
GC Content (%)	$(181 + 177) / 628 \times 100\%$	57.01%
AT Content (%)	$(142 + 128) / 628 \times 100\%$	42.99%

Biological Interpretation (135 words):

The retrieved sequence (NCBI Accession: NM_000518.5) represents the full-length mature mRNA transcript of the human beta-globin (*HBB*) gene, consisting of 628 base pairs. Quantitative sequence analysis reveals a GC content of 57.01% and an AT content of 42.99%. In mammalian genomes, a GC content exceeding 50% is a hallmark of exonic coding regions and functional mRNA transcripts, contrasting with GC-depleted non-coding introns and intergenic regions. The elevated GC content provides structural rigidity and thermal stability to mRNA secondary folds, protecting the transcript against enzymatic degradation and facilitating rapid ribosomal translation in erythroid cells. Furthermore, the balanced base distribution reflects evolutionary codon optimization required for efficient synthesis of functional hemoglobin subunits.

Q6. Critical Thinking

Can two DNA sequences have the same length but different GC contents? Explain your answer with a suitable biological example.

Answer: Yes, absolutely. DNA sequence length (the total count of nucleotides) is completely independent of sequence composition (the relative proportion of individual bases).

Biological Explanation & Example:

Consider two gene fragments of identical length (1,000 bp) isolated from organisms adapted to vastly different ecological niches:

Feature / Property	Sequence 1 (<i>Thermus aquaticus</i>)	Sequence 2 (<i>Plasmodium falciparum</i>)
Sequence Length	1,000 bp	1,000 bp
G + C Base Count	680 bp	200 bp
A + T Base Count	320 bp	800 bp
GC Content (%)	68.0% GC	20.0% GC
Environmental Adaptation	Hyperthermophile (hot springs, high thermal stability)	Eukaryotic intracellular parasite (AT-biased genome)

Conclusion: Although both DNA fragments have the exact same length (1,000 bp), evolutionary pressures, environmental thermal requirements, codon selection bias, and distinct DNA replication enzymes lead to radically different GC percentages.